Ajay Devidas Wanekar

M.Tech Energy Science and Technology - TA

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SUMMARY

M.Tech student at IIT Hyderabad (B.E. Electrical Engineering) with 2+ years of research experience in data analysis and Machine Learning. Experienced in modelling a plasma process from limited experimental data, root-cause and feature-importance analysis, small-object detection in noisy images, and leading a team of 5 to take a customer-facing AI product from idea to prototype in the IIT Hyderabad Build Program. Python ML scripting, statistical validation, two IEEE publications, and clear technical writing as a Teaching Assistant.

Degree	University/Institute	Year	CGPA/ Marks(%)
M.Tech Energy Science and Technology - TA	IIT Hyderabad	2027	6.8
B.E. Electrical Engineering	Savitribai Phule Pune University	2022	7.28
XII (Maharashtra State Board)	Vasantrao Naik Jr. College of Science and Arts, Cidco, Ch. Sambhajinagar, Maharashtra	2017	67.38%
X (Maharashtra State Board)	Maharashtra Public School, DTP, Ch. Sambhajinagar	2014	83.00%

WORK EXPERIENCE

Project Associate | LFOVIA Lab, IIT Hyderabad Sep 2024 – Jul 2025

Evaluated and fine-tuned Vision-Language Models (Florence-2, Llama 3.2 Vision, Qwen2-VL) for detection, segmentation, and text recognition, and benchmarked them against task-specific models using precision, recall, and IoU in Python analysis pipelines.

Junior Research Assistant | TIHAN, IIT Hyderabad: Nov 2023 – Jul 2024

Built object detection and tracking (YOLO, DeepSORT) and curated near-infrared and sonar datasets with sensor-specific preprocessing, annotation, and data-quality checks for reliable detection of small objects in noisy, low-contrast images, resulting in an IEEE conference publication.

Research Intern | WINET Lab, IIT Hyderabad: Jul 2023 – Nov 2023

Developed sonar-based underwater object-detection datasets with annotation, Python preprocessing, data augmentation, and quality checks, published at IEEE i-COSTE 2023.

THESIS PROJECT

Machine Learning for Non-Oxidative Methane Coupling in a DBD Plasma Reactor

- Modelled six process outputs (CH_4 conversion, H_2 and C_2–C_4 selectivities) from reactor operating conditions with only 24 experiments, benchmarking Linear Regression, Random Forest, XGBoost, and Gaussian Process Regression with Leave-One-Out Cross-Validation (cross-validated R^2 up to 0.996).
- Ranked the operating conditions driving each output with feature-importance analysis, studied extrapolation and confounding limits, adapted the RRSI method to verify gains exceed experimental noise, and identified Bayesian Optimization for experiment-guided process optimization.

PROJECTS

Multi-Tenant Conversational AI Platform, Team Lead (IIT Hyderabad Build Program):

- Led a team of 5 from idea to working prototype: defined the problem statement and customer use cases, set priorities, split tasks across the team, and ran several workstreams (voice, chat, email, retrieval) in parallel.
- Delivered a customer-facing multi-tenant generative-AI platform, configured it for each customer's use case, and resolved issues as they arose, the same cycle as new product introduction and customer support.
- Designed a hybrid semantic + keyword retrieval pipeline with LLM-based intent classification that improved answer relevance by 18%; trained and tested speech-to-text and text-to-speech models.

Robust Classification Under Shifted Conditions: Asbestos-Cement Roofs (open source, GitHub):

- Classified roofs from 8-band satellite spectra in a region the model had never seen: 95.2% accuracy (F1 0.955) vs 90.2% for the dataset authors' best model, by engineering brightness-robust features and re-fitting the decision threshold to the new region's unlabelled data (Gaussian Mixture Model).

TECHNICAL SKILLS

Programming & Tools: Python, SQL, C++, Linux, Pandas, NumPy, Scikit-learn, XGBoost, PyTorch, TensorFlow, OpenCV, Matplotlib

Data Analysis & Statistics: Exploratory Data Analysis, Root-Cause and Feature-Importance Analysis, Regression, Random Forest, SVM, Gaussian Processes, Cross-Validation (LOOCV), Statistical Validation, Bayesian Optimization

Process & Engineering: Electrical Engineering Fundamentals, Plasma (DBD) Process Modelling, Experimental Data Analysis, Process Optimization

Inspection & AI: Object Detection (YOLO), Image Preprocessing, Data Annotation, Vision-Language Models, LLMs, RAG

Leadership & Communication: Team Leadership (team of 5), Problem Definition, Prioritization, Multi-Tasking, Customer Use-Case Development, Technical Writing (2 IEEE papers), Teaching Assistant, Documentation

CONFERENCE PUBLICATIONS

[1] Wanekar A. D., Mannam N. P. B., Rajalakshmi P. “Novel Approach to Underwater Object Detection Using Sonar Sensors for Autonomous Underwater Vehicles (AUVs).” IEEE i-COSTE 2023, pp. 1–5.

[2] Wanekar A. D., Mannam N. P. B., Rajalakshmi P. “Object Detection and Classification for Autonomous Surface Vehicles (ASVs) Through Near-Infrared Imaging.” IEEE i-COSTE 2023, pp. 1–6.

HACKATHONS, ACHIEVEMENTS & LANGUAGES

- Top 10 All India - Winner, Meta x Pragati AI Hackathon.
- AI and X Hackathon, IIT Hyderabad participant

Languages: English, Hindi, Marathi, Japanese (Basic)